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ASSIGNMENT(POE PART1) API

Convention: Every route starts with API

Value	Meaning
None (public)	No token required.
Any (logged in)	Valid JWT; Organiser or Participant.
Organiser	JWT plus role claim <code>Organiser</code> .
Participant	JWT plus role claim <code>Participant</code> .

Failure responses are listed in Expected Response so Part 2 can return consistent HTTP codes (Fielding, 2000; W3Schools, n.d.a).

Authentication

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	<code>/api/auth/register</code>	Creates a new RaceDay account as Organiser or Participant.	None (public)	<pre>{ email, password, firstName, lastName, role, phone, clubId, dateOfBirth }</pre>	201 Created with user profile (no password). 400 Bad Request invalid fields or unknown role. 409 Conflict email already registered.
POST	<code>/api/auth/login</code>	Authenticates a user and returns a JWT.	None (public)	<pre>{ email, password }</pre>	200 OK { token, expiresAt, role, userId }. 400 Bad Request missing fields. 401 Unauthorized unknown email or wrong password.

User Profile

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/users/me	Returns the authenticated user's profile.	Any (logged in)	None	200 OK profile object. 401 Unauthorized missing/invalid token.
PUT	/api/users/me	Updates name, phone, club and date of birth (not role).	Any (logged in)	{ firstName, lastName, phone, clubId, dateOfBirth }	200 OK updated profile. 400 Bad Request validation errors. 401 Unauthorized. 404 Not Found user row missing.
GET	/api/users/{id}	Organiser lookup of a participant (enrolment support).	Organiser	None	200 OK public profile fields. 401/403. 404 Not Found.

Events

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/events	Lists published upcoming events for browsing.	None (public)	Query: sportType, province, from, to	200 OK event array (may be empty). 400 Bad Request invalid filter.
GET	/api/events/{id}	Returns one event with categories, route summary and latest weather.	None (public)	None	200 OK event detail. 404 Not Found.
POST	/api/events	Creates an event owned by the authenticated organiser.	Organiser	{ name, description, sportType, venue, city, province, startDateTime, endDateTime, status, routeSummary }	201 Created. 400 Bad Request. 401/403.
PUT	/api/events/{id}	Updates an event the organiser owns.	Organiser	Same fields as create	200 OK. 400. 401/403 (including not the owner). 404 Not Found.
DELETE	/api/events/{id}	Deletes an event the organiser owns if no confirmed enrolments block it.	Organiser	None	204 No Content. 401/403. 404. 409 Conflict enrolments exist.

Categories

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/events/{eventId}/categories	Lists categories (distances and fees) for an event.	None (public)	None	200 OK array. 404 Not Found event missing.
POST	/api/events/{eventId}/categories	Adds a category to an event.	Organiser	{ name, distanceKm, entryFeeZar, maxParticipants, minAge, maxAge }	201 Created. 400. 401/403. 404 event missing.
PUT	/api/categories/{id}	Updates a category.	Organiser	Same fields as create	200 OK. 400. 401/403. 404.
DELETE	/api/categories/{id}	Removes a category that has no enrolments.	Organiser	None	204 No Content. 401/403. 404. 409 Conflict enrolments exist.

Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	/api/enrolments	Participant enters an event by selecting a category.	Participant	{ eventId, categoryId }	201 Created enrolment (bib assigned). 400 category not on that event. 401/403. 404 event/category missing. 409 Conflict already entered.
GET	/api/enrolments/me	Participant lists their own enrolments and linked events.	Participant	None	200 OK array. 401/403.
GET	/api/events/{id}/enrolments	Organiser views all enrolments for an event they own.	Organiser	None	200 OK array. 401/403. 404.
PATCH	/api/enrolments/{id}	Participant withdraws, or organiser updates bib/status.	Participant or Organiser	{ status } or { bibNumber, status }	200 OK. 400. 401/403 not owner of enrolment/event. 404. 409 illegal status change.

Results

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	/api/results	Organiser captures a result for an enrolment.	Organiser	{ enrolmentId, finishTime, positionOverall, resultStatus }	201 Created. 400. 401/403. 404. 409 Conflict result already exists.
PUT	/api/results/{id}	Organiser corrects a captured result.	Organiser	{ finishTime, positionOverall, resultStatus }	200 OK. 400. 401/403. 404.
GET	/api/results/me	Participant views personal result history.	Participant	None	200 OK array. 401/403.
GET	/api/events/{id}/results	Published results list for an event.	None (public)	None	200 OK array. 404 event missing.

Route Information

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/events/{id}/routes	Returns route notes and GPX/blob URLs for an event.	None (public)	None	200 OK array. 404.
POST	/api/events/{id}/routes	Organiser adds a route record (Part 3 will upload GPX to Azure Blob).	Organiser	{ name, distanceKm, elevationGainM, mapNotes, gpxUrl }	201 Created. 400. 401/403. 404.

Live Weather Race Day (Race-Day Preparation)

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/events/{id}/weather	Returns the latest stored snapshot; Part 3 may refresh from a live weather API.	None (public)	None	200 OK snapshot. 404 event or snapshot missing.
POST	/api/events/{id}/weather	Organiser (or system job) stores a weather snapshot.	Organiser	{ temperatureC, condition, windKph, humidityPercent, source }	201 Created. 400. 401/403. 404.

Clubs (supporting profile data)

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/clubs	Lists clubs for registration drop-downs.	None (public)	None	200 OK array

Role-based access control (enforced in Part 2, reflected in Part 3 MVC)

Capability	Organiser	Participant	Public
Register / login	Yes	Yes	Register/login only
Browse events, routes, weather, published results	Yes	Yes	Yes
Create / update / delete events and categories	Yes	No	No
Capture and edit results	Yes	No	No
View all enrolments for owned events	Yes	No	No
Enter an event (choose category)	No	Yes	No
View own enrolments and own results	No	Yes	No

Unauthorised calls must return 401 (no token) or 403 (wrong role), not a silent 404, so the MVC client in Part 3 can show a clear message (Sandhu et al., 1996).